Can we find delivery types without labels?

K-means clustering, on real ball-tracking data

The cricket question

— A tracker prints speed, swing, seam deviation for every ball.

— Could you tell fast bowler from spinner on those numbers alone?

— Is 'pace or spin' even the only real split?

The data

— 194,592 deliveries, 7 physical measurements per ball.

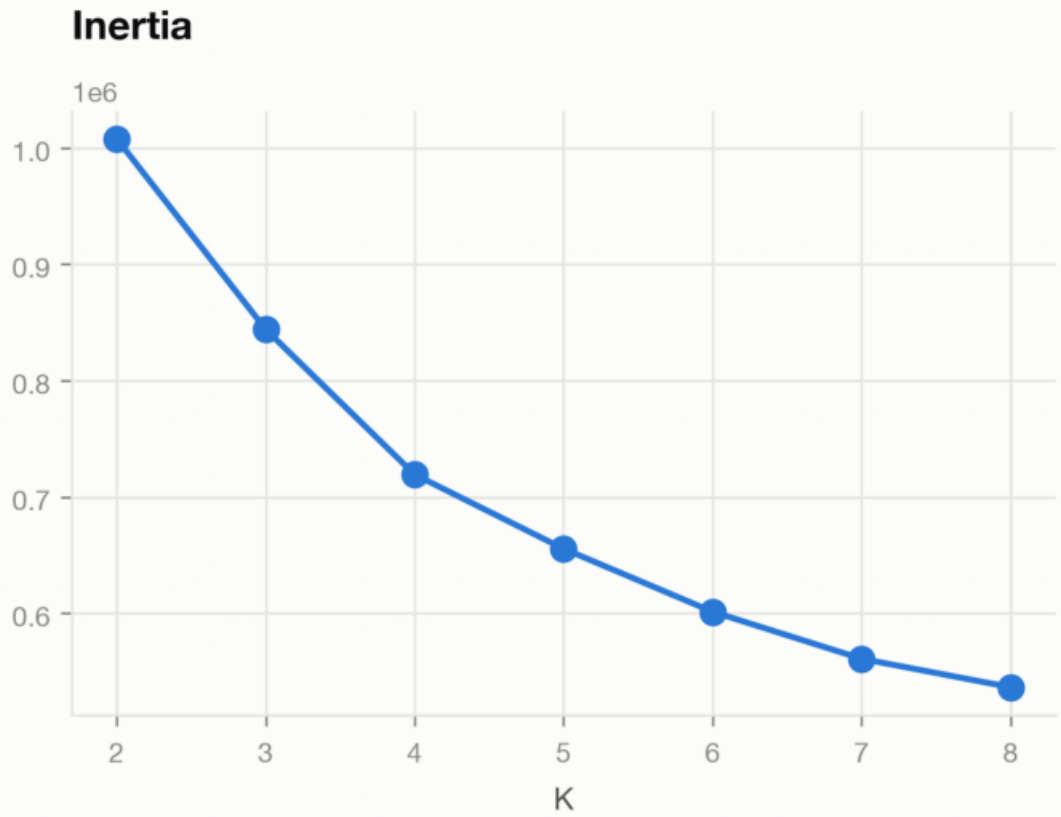
— No player identity — this clusters deliveries, not bowlers.

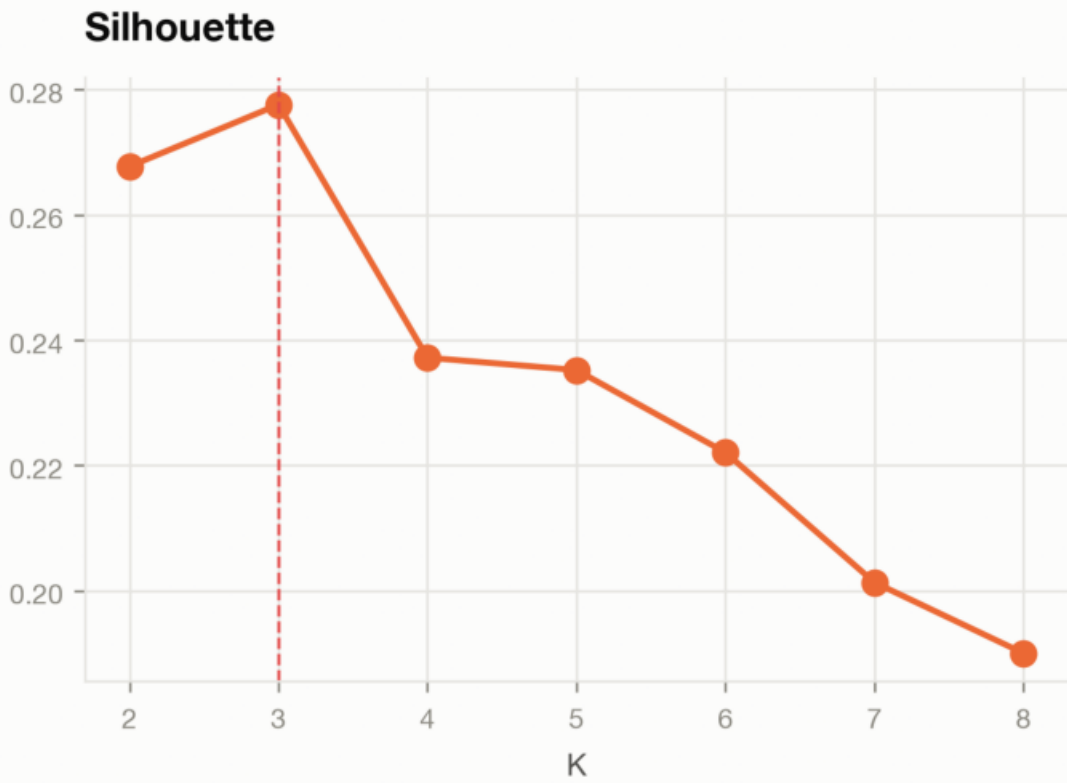
— bowling_type (pace/spin) is held out — used only to grade clusters.

The naive answer

- Cricket already has a 2-bucket answer: pace or spin.
- Assigned from the run-up, not the ball's physics.
- Any clustering should recover this before it earns credit for more.

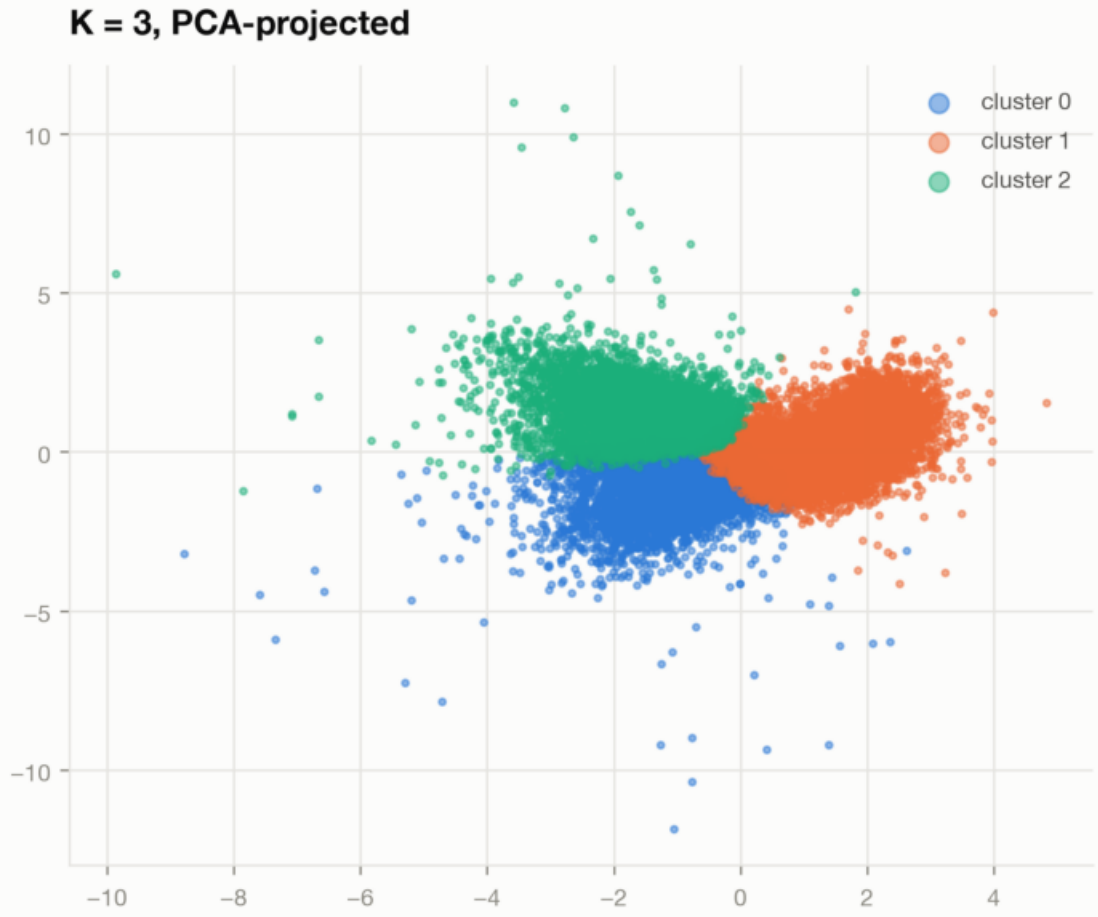
Choosing K: elbow + silhouette





Silhouette peaks at K=3

K = 3, projected to 2D



What the clusters found

cluster	share	pace %	spin %	wicket rate
mixed	24.9%	42.2%	57.8%	5.3%
genuine pace	51.8%	97.1%	2.9%	5.0%
genuine spin	23.3%	6.7%	93.3%	4.7%

pace/spin purity was withheld from training

What it gets wrong

— Only one label (pace/spin) exists to validate against.

— K chosen by silhouette on a 10k-row subsample, for speed.

— Standardising treats line & length as equal to speed — they aren't, cricket-wise.

— A from-scratch single-init run misses scikit-learn's answer (ARI 0.396) — n_init exists for a reason.

Next: compressing these features with PCA

cricketml